Scaling Up Visual and Vision-Language Representation Learning With Noisy Text Supervision

https://arxiv.org/abs/2102.05918

0. Introduction

- 웹에는 이미지와 텍스트가 대규모로 존재하지만 노이즈가 많음
- 기존 모델은 대부분 정제된 라벨 데이터에 의존
- 본 연구는 대규모 noisy image-text 데이터를 활용해 vision-language representation을 학습하는 방법 제안
- 약 1.8B image-text pairs로 모델을 학습하여 zero-shot 성능 향상

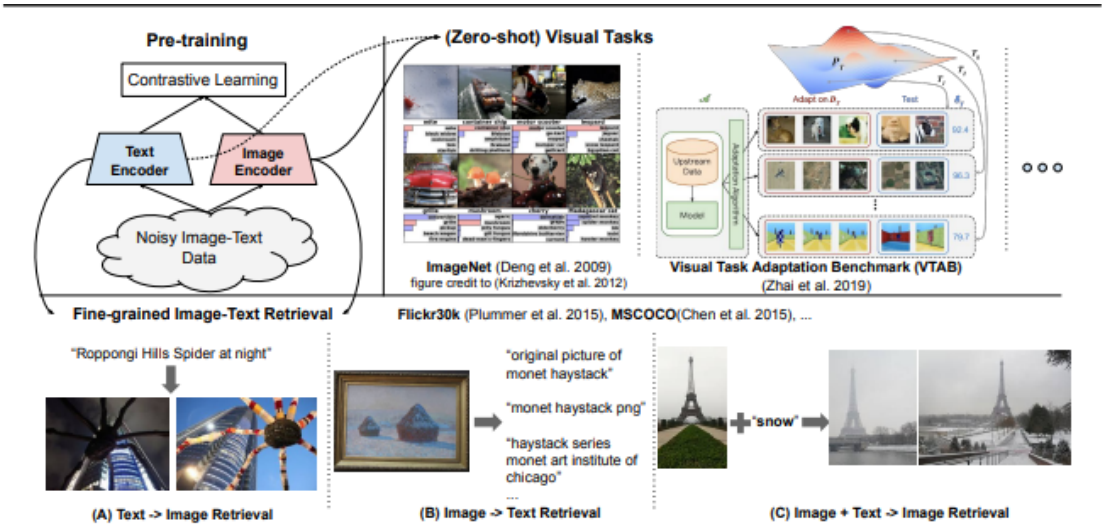
1. Overview

- 이미지와 텍스트를 각각 encoder로 변환 후 **공통 embedding 공간에서 정렬**
- Image encoder : EfficientNet
- Text encoder : Transformer 기반 모델
- **contrastive learning**을 사용하여 image-text alignment 학습

2. Challenges

- 웹 데이터의 텍스트 노이즈 문제
- 이미지와 텍스트 의미 불일치
- 대규모 데이터 학습 시 연산 비용 증가

3. Method



- 약 **1.8B image-text pairs 데이터셋 사용**
- image encoder와 text encoder를 통해 embedding 생성
- **contrastive loss**를 사용하여 positive pair 유사도 증가
- 대규모 batch와 distributed training 적용

4. Experiments

- 평가 태스크
  - image classification
  - image-text retrieval
  - zero-shot classification
- 비교 모델
  - supervised ImageNet 모델
  - Learning Transferable Visual Models From Natural Language Supervision (CLIP)

5. Results

		Flickr30K (1K test set)						MSCOCO (5K test set)					
		image → text			text → image			image → text			text → image		
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
Zero-shot	ImageBERT	70.7	90.2	94.0	54.3	79.6	87.5	44.0	71.2	80.4	32.3	59.0	70.2
	UNITER	83.6	95.7	97.7	68.7	89.2	93.9	-	-	-	-	-	-
	CLIP	88.0	98.7	99.4	68.7	90.6	95.2	58.4	81.5	88.1	37.8	62.4	72.2
	ALIGN	**88.6**	**98.7**	**99.7**	**75.7**	**93.8**	**96.8**	**58.6**	**83.0**	**89.7**	**45.6**	**69.8**	**78.6**
Fine-tuned	GPO	88.7	98.9	99.8	76.1	94.5	97.1	68.1	90.2	-	52.7	80.2	-
	UNITER	87.3	98.0	99.2	75.6	94.1	96.8	65.7	88.6	93.8	52.9	79.9	88.0
	ERNIE-ViL	88.1	98.0	99.2	76.7	93.6	96.4	-	-	-	-	-	-
	VILLA	87.9	97.5	98.8	76.3	94.2	96.8	-	-	-	-	-	-
	Oscar	-	-	-	-	-	-	73.5	92.2	96.0	57.5	82.8	**89.8**
	ALIGN	**95.3**	**99.8**	**100.0**	**84.9**	**97.4**	**98.6**	**77.0**	**93.5**	**96.9**	**59.9**	**83.3**	**89.8**

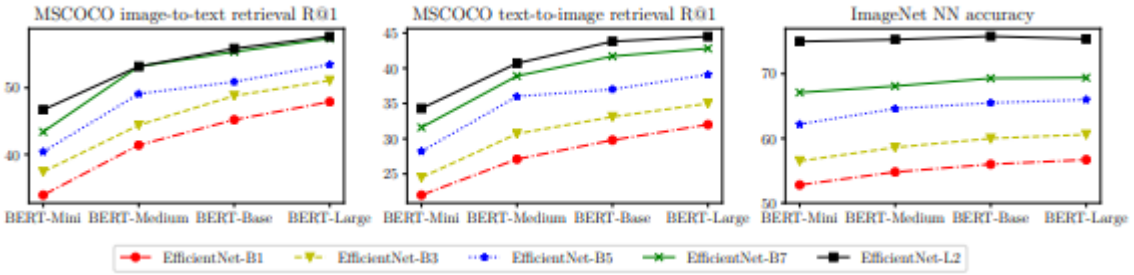
	image → text			text → image			text → text			image → image		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
VSE++	43.1	74.3	84.2	32.5	62.7	75.4	38.7	62.3	72.2	36.4	70.4	81.3
VSRN	52.4	81.9	90.0	40.1	71.1	81.5	41.0	64.8	74.5	44.2	76.7	86.2
DE_{I2T}	53.9	82.7	91.2	39.8	70.2	80.9	26.0	47.1	57.5	38.3	74.1	85.0
$DE_{T2T+I2T}$	55.9	84.2	91.8	41.7	72.3	83.0	42.4	64.9	74.0	38.5	73.6	84.9
ALIGN	**78.1**	**94.3**	**97.4**	**61.8**	**84.9**	**91.1**	**45.4**	**66.8**	**75.2**	**49.4**	**81.4**	**89.1**

Model	**STS** avg ± std	**SIS** avg ± std	**SITS** avg ± std	**Mean Avg**
VSE++	**74.4±0.4**	73.3±0.9	55.2±1.5	67.6
VSRN	73.0±0.4	70.1±1.0	60.4±1.3	67.8
DE_{I2T}	50.9±0.6	**81.3±0.7**	61.6±1.4	64.6
$DE_{T2T+I2T}$	74.2±0.4	74.5±0.9	61.9±1.3	70.2
ALIGN	72.9±0.4	77.2±0.8	**67.6±1.2**	**72.6**

Model	ImageNet	ImageNet-R	ImageNet-A	ImageNet-V2
CLIP	76.2	88.9	**77.2**	**70.1**
ALIGN	**76.4**	**92.2**	75.8	**70.1**

Model	All tasks	Natural	Specialized	Structured
Bit-L	78.72	-	-	-
ALIGN	**79.99±0.15**	83.38	87.56	73.25

Model (backbone)	Acc@1 w/ frozen features	Acc@1	Acc@5
WSL (ResNeXt-101 32x48d)	83.6	85.4	97.6
CLIP (ViT-L/14)	85.4	-	-
BiT (ResNet152 x 4)	-	87.54	98.46
NoisyStudent (EfficientNet-L2)	-	88.4	98.7
ViT (ViT-H/14)	-	88.55	-
Meta-Pseudo-Labels (EfficientNet-L2)	-	**90.2**	**98.8**
ALIGN (EfficientNet-L2)	**85.5**	88.64	98.67



- zero-shot classification에서 높은 성능 달성
- 대규모 noisy 데이터에서도 **강한 일반화 성능 확인**
- supervised 모델과 경쟁 가능한 결과

6. Insight

- **데이터 규모가 모델 성능에 매우 큰 영향**
- noisy 웹 데이터도 대규모이면 유용한 representation 학습 가능
- vision-language foundation model의 가능성을 보여준 연구